ModelIt!

Teacher's Guide

What Do Animals Need to Survive?

The Science of Animal Needs

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-LS1-1

Use materials to design a solution to a human problem by mimicking how plants and animals use their external parts to help them survive, grow, and meet their needs.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students build a model habitat showing how resources connect to animal health and survival.
Disciplinary Core Idea	**LS1.A Structure and Function** — All organisms have external parts that they use to perform daily functions including survival, growth, and reproduction.
Crosscutting Concept	**Cause and Effect** — Students explore how having or lacking resources causes animals to be healthy or unhealthy.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every animal on Earth needs certain things to survive. From tiny ants to giant elephants, all animals share the same basic needs.

The Four Basic Needs

All animals need four things to stay alive: food gives them energy to move, grow, and stay warm; water keeps their bodies working properly since all living things are mostly made of water; air provides oxygen that every animal needs to breathe; and shelter protects them from bad weather, extreme temperatures, and predators. If any one of these needs is missing, the animal will struggle to survive.

Habitats Provide What Animals Need

A habitat is the natural home of an animal — the place where it can find all four of its basic needs. A pond is a great habitat for a frog because it has water to swim in, insects to eat, air to breathe, and lily pads and mud for shelter. A desert is perfect for a lizard because it has insects to eat, small water sources, air, and rocks for shelter. When a habitat is damaged or changed, animals may lose the resources they need.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify resources available as something that depends on the habitat (external) and animal health as something that changes based on those resources (internal).

E — Establish Relationships
Students discover that when resources go up, animal health goes up — a positive connection.

V — Visualize & Model
Students build a simple model showing how the amount of resources affects animal health.

E — Evaluate Outcomes
Students test their model by comparing what happens with plenty of resources versus very few resources.

R — Revise & Extend
Students reflect on how their model explains why some animals thrive in certain habitats and struggle in others.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Think about your pet or an animal you know. What does it need every single day? Let us make a list!"*

Watch For

Create a class list of things animals need. Accept all answers and revisit at the end to see which ones are correct.

Slide 2: What Will We Learn?

3 min

Say: *"Today we are going to learn the four things every animal needs to survive. We will also learn what a habitat is!"*

Watch For

Introduce vocabulary: survive, shelter, habitat. Use pictures and gestures for each word.

Slide 3: Let's Be Scientists!

3 min

Say: *"What do animals need to stay alive and healthy? Do all animals need the same things?"*

Watch For

Think-pair-share. Students discuss what they think animals need most. Collect ideas.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Resources Available is how much food, water, and shelter is around. Animal Health is how strong and healthy the animal is."*

Watch For

Show pictures of a well-fed animal vs. a thin animal. Ask: What is different? Why?

Slide 5: How Are They Connected?

5 min

Say: *"When there are MORE resources, what happens to the animal? Does it get healthier or weaker?"*

Watch For

Students use sentence frame: When resources go up, animal health goes up. Draw the positive arrow together.

Slide 6: What Did We Find Out? 5 min

Say: *"We discovered that all animals need food, water, air, and shelter. When they have these things, they are healthy!"*

Watch For

Review the class list from Slide 1. Check off which items were correct. Add any that were missing.

Slide 7: STEM Challenge 10 min

Say: *"Time to build a home for an animal! Make sure your habitat has food, water, and shelter."*

Watch For

Distribute shoeboxes and craft materials. Students choose an animal and build its habitat, labeling the four needs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Resources Available
- **INTERNAL:** Animal Health

Relationships

- **Resources Available to Animal Health:** POSITIVE (+) — When more resources like food, water, and shelter are available, the animal stays healthier and stronger. Less resources means the animal gets weaker.

Key Discoveries

- All animals need food, water, air, and shelter to survive
- When animals have plenty of resources, they are healthy and strong
- When resources are scarce, animals get weak and may not survive
- Different animals find their needs in different ways

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Animals only need food to survive"	Food is important but it is only ONE of four things animals need. Without water, animals would dry out. Without air, they could not breathe. Without shelter, they would have no protection from weather or predators. All four needs must be met.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All animals need the same kind of food"	Different animals eat very different things. Lions eat meat, rabbits eat plants, and bears eat both. But ALL animals need SOME kind of food for energy. The type of food changes, but the need for food stays the same for every animal.